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Class and Batch	TE Computer Science and Engineering - Batch B1
Date	17/09/2025
Lab.	CNS
Experiment No.	3

Aim:	Asymmetric Key Cryptography
Objective:	Implement the RSA algorithm.
Problem Statement:	The experiment uses asymmetric and symmetric cryptosystems to provide two security services. First, this experiment covers key distribution problems using asymmetric cryptosystems. The asymmetric Cryptosystems allow sharing secret keys between sender and receiver through a third-party entity. Second, symmetric cryptosystem covers sending large message from the sender to the receiver using the secret key shared in the first step.
Theory:	Public-key cryptography, or asymmetric cryptosystem, is the field of cryptographic systems that use pairs of related keys. Each key pair consists of a public key and a corresponding private key. Key pairs are generated with cryptographic algorithms based on mathematical problems termed one-way functions (i.e. discrete logarithm problem). Security of public-key cryptography depends on keeping the private key secret; the public key can be available, and it does not compromise security.
Code:	Sender.py: <pre>import socket, pickle, hashlib from rsa_utils import generate_keys, rsa_encrypt, xor_encrypt host = "0.tcp.in.ngrok.io" port = 18832</pre>

```
name = input("Sender name: ")
```

```
pubS, privS = generate_keys()
```

```
s = socket.socket(); s.connect((host, port))
```

```
s.send(pickle.dumps(("register", name, pubS)))
```

```
print("Third Party:", pickle.loads(s.recv(4096)))
```

```
s.close()
```

```
recv_name = input("Enter receiver name: ")
```

```
s = socket.socket(); s.connect((host, port))
```

```
s.send(pickle.dumps(("get", recv_name, pubS)))
```

```
recv_pub = pickle.loads(s.recv(4096))
```

```
s.close()
```

```
if not recv_pub:
```

```
    print("Receiver not registered!")
```

```
    exit()
```

```
sym_key = 42
```

```
msg = ''
```

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```
'''
hash_val = hashlib.sha256(msg.encode()).hexdigest()

enc_key = rsa_encrypt(sym_key, recv_pub)

cipher_msg = xor_encrypt(msg, sym_key)

r = socket.socket()
r.connect(('127.0.0.1', 3333))
r.send(pickle.dumps((enc_key, cipher_msg, hash_val)))
print("Message sent!")
r.close()
```

Receiver.py:

```
import socket, pickle, hashlib
from rsa_utils import generate_keys, rsa_decrypt, xor_decrypt

host = "127.0.0.1"
port = 3333

name = input("Receiver name: ")

pubR, privR = generate_keys()

s = socket.socket(); s.connect(("0.tcp.in.ngrok.io", 18832))
```

```

s.send(pickle.dumps(("register", name, pubR)))
print("Third Party:", pickle.loads(s.recv(4096)))
s.close()

srv = socket.socket()
srv.bind((host, port))
srv.listen(1)
print("Receiver listening on", port)

while True:
    conn, addr = srv.accept()
    print("New host joined")
    data = pickle.loads(conn.recv(10000000))
    enc_key, cipher_msg, hash_val = data
    print(enc_key)
    print(cipher_msg)

    sym_key = int(rsa_decrypt(enc_key, privR))

    plain = xor_decrypt(cipher_msg, sym_key)
    recv_hash = hashlib.sha256(plain.encode()).hexdigest()
    print("Received message:", plain)

    if(hash_val == recv_hash):
        print("Received data matches the original data by hash
value")
    else:
        print("Received data does not match the original data by hash
value")

    conn.close()

```

Rsa_utils.py

```

import random

def power(base, expo, m):
    res = 1
    base = base % m
    while expo > 0:
        if expo & 1:

```

```

        res = (res * base) % m
        base = (base * base) % m
        expo = expo // 2
    return res

def isPrime(n):
    if n < 2:
        return False
    for i in range(2, int(n ** 0.5) + 1):
        if n % i == 0:
            return False
    return True

def generate_prime(start=10000, end=30000):
    while True:
        p = random.randint(start, end)
        if isPrime(p):
            return p

def gcd(a, b):
    if a < b:
        a, b = b, a
    while b != 0:
        a, b = b, a % b
    return a

def modInverse(e, phi):
    for d in range(2, phi):
        if (e * d) % phi == 1:
            return d
    return -1

def generate_keys():
    p = generate_prime()
    q = generate_prime()
    n = p * q
    phi = (p - 1) * (q - 1)
    for e in range(2, phi):
        if gcd(e, phi) == 1:
            break
    d = modInverse(e, phi)

```



```

        return (e, n), (d, n)

def rsa_encrypt(m, pub):
    e, n = pub
    return power(m, e, n)

def rsa_decrypt(c, priv):
    d, n = priv
    return power(c, d, n)

def xor_encrypt(text, key):
    return ''.join(chr(ord(c) ^ key) for c in text)

def xor_decrypt(cipher, key):
    return ''.join(chr(ord(c) ^ key) for c in cipher)

```

ThirdParty.py:

```

import socket, threading, pickle

HOST = "0.0.0.0"
PORT = 4444
store = {}

def handle_client(conn):
    req = conn.recv(4096)
    if not req: return
    action, name, key = pickle.loads(req)
    if action == "register":
        store[name] = key
        conn.send(pickle.dumps("Registered " + name))
        print(name, "-", key)
    elif action == "get":
        conn.send(pickle.dumps(store.get(name, None)))
    conn.close()

s = socket.socket()
s.bind((HOST, PORT))
s.listen(5)
print("Third Party running on", PORT)

```

```
while True:
    c, _ = s.accept()
    threading.Thread(target=handle_client, args=(c,)).start()
```

Output:

```
OUTPUT DEBUG CONSOLE TERMINAL PORTS OUTLINES PROBLEMS
[env:] aryanah@AryanMacbook-Air-2 OS: macOS python3 sender.py
Sender name: Alice
Port: 3971 Registered alive
Receiver name: Bob
Message sent!
[env:] aryanah@AryanMacbook-Air-2 OS: macOS
```

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```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\saran\Documents\Assignments\~
TE Folder~\CNS> python .\ThirdParty.py
Third Party running on 4444
Alpha - (3, 31351)
Gamma - (3, 25651)

PS C:\Users\saran\Documents\Assignments\~
TE Folder~\CNS> python .\sender.py
Sender name: Alpha
Third Party: Registered Alpha
Enter receiver name: Gamma
Enter message: hello
Message sent!

PS C:\Users\saran\Documents\Assignments\~
TE Folder~\CNS>

PS C:\Users\saran\Documents\Assignments\~
TE Folder~\CNS> python .\receiver.py
Receiver name: Gamma
Third Party: Registered Gamma
Receiver listening on 3333
New host joined
22786
BOFFE
Received message: hello
```